

CSE5026: Introduction and Frontiers of Cognitive Science

Emotion and Affective Computation

Yang Xu 徐炆

Department of Computer Science and Engineering
Southern University of Science and Technology

Overview

- Emotion Theories
- Affective Computation

Emotion 情绪

Source: <https://en.wikipedia.org/wiki/Emotion>

- Emotion: **physical** and **mental** states brought on by neurophysiological changes, variously associated with thoughts, feelings, behavioral responses, and a degree of pleasure or displeasure.
- Darwin argued that emotional expressions have evolutionary origins and can serve adaptive and communicative functions.
(The expression of Emotions In Man and Animals, 1872)

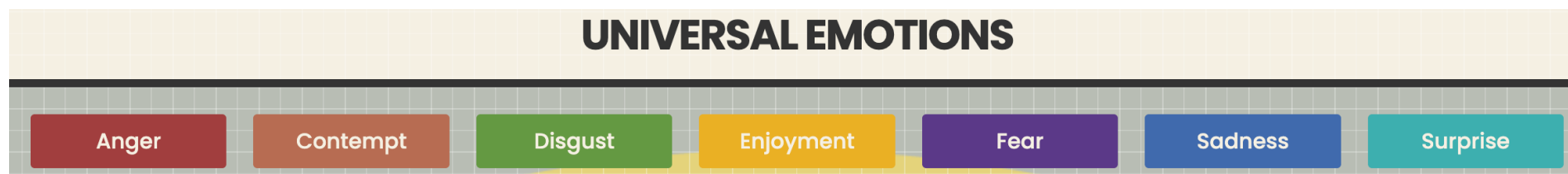


illustration from Darwin's book

情绪的分类 Classification

Source: <https://www.paulekman.com/about/paul-ekman/>

- American psychologist Paul Ekman's basic emotions theory
 - Early theory: 6 basic emotions:
anger, disgust, fear, happiness, sadness and surprise
愤怒, 厌恶, 恐惧, 高兴, 悲伤, 惊讶
 - His later research shows evidence of a third emotion, contempt (轻蔑)
- Now on [P. Ekman's website](https://www.paulekman.com/about/paul-ekman/) :



我国文化中有关情绪的理论

Source: <https://zh.wikipedia.org/zh-cn/七情>

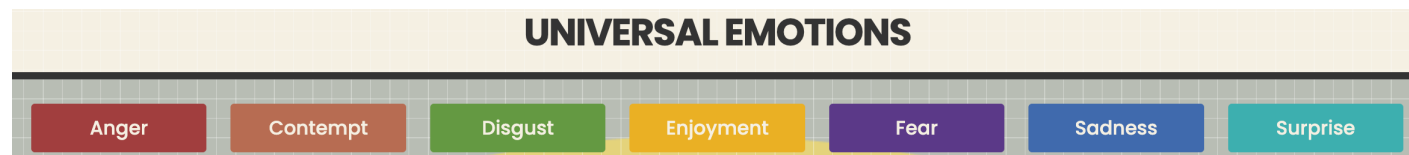
- “七情”

- 佛教： 喜，怒，忧，惧，爱，憎，欲

- 儒家： ① 喜，怒，哀，惧，爱，恶，欲
② 喜，怒，哀，**乐**，爱，恶，欲

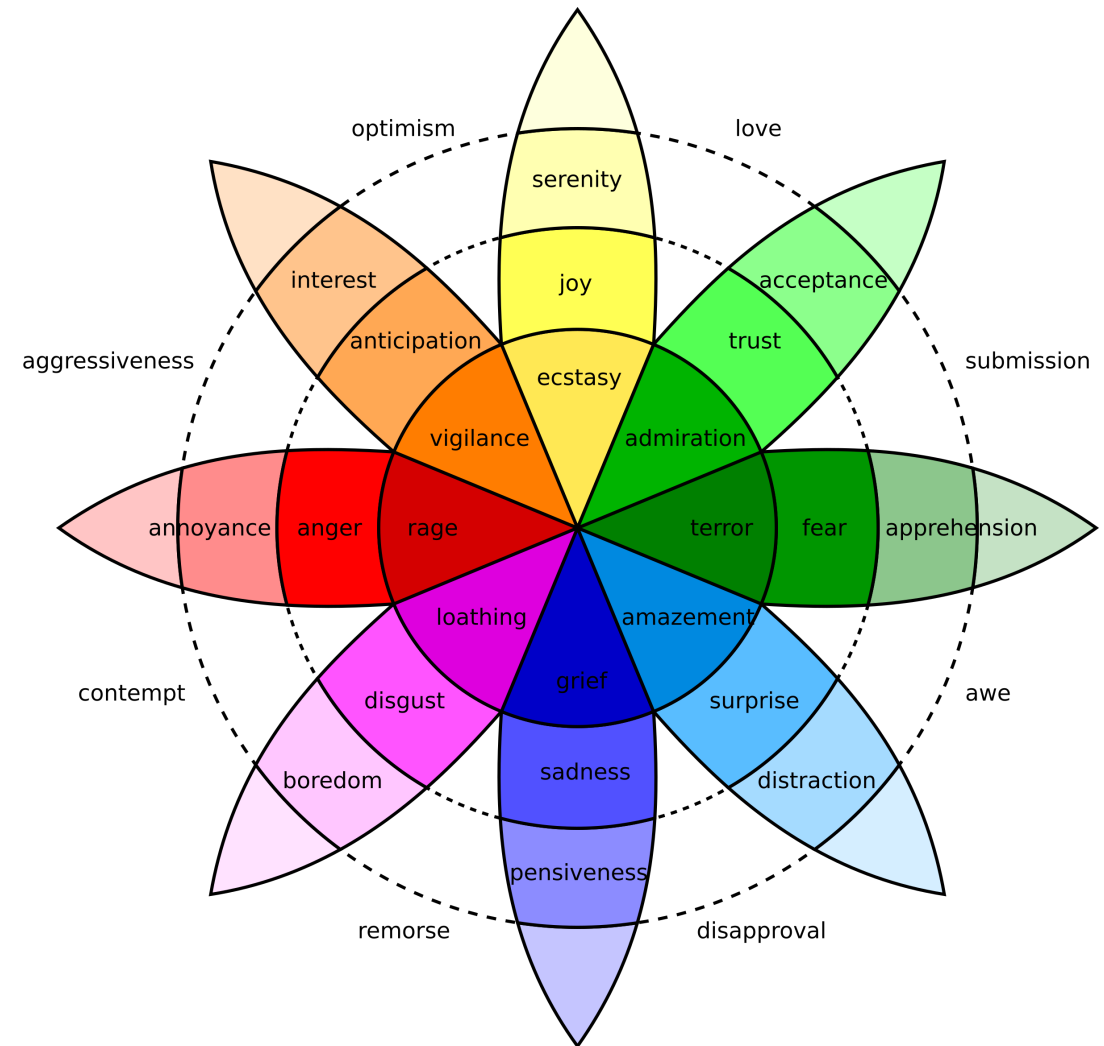
- 中医： 喜，怒，忧，思，悲，恐，惊
(道家)

joy 乐 ≠ 喜 happy



Other Theories

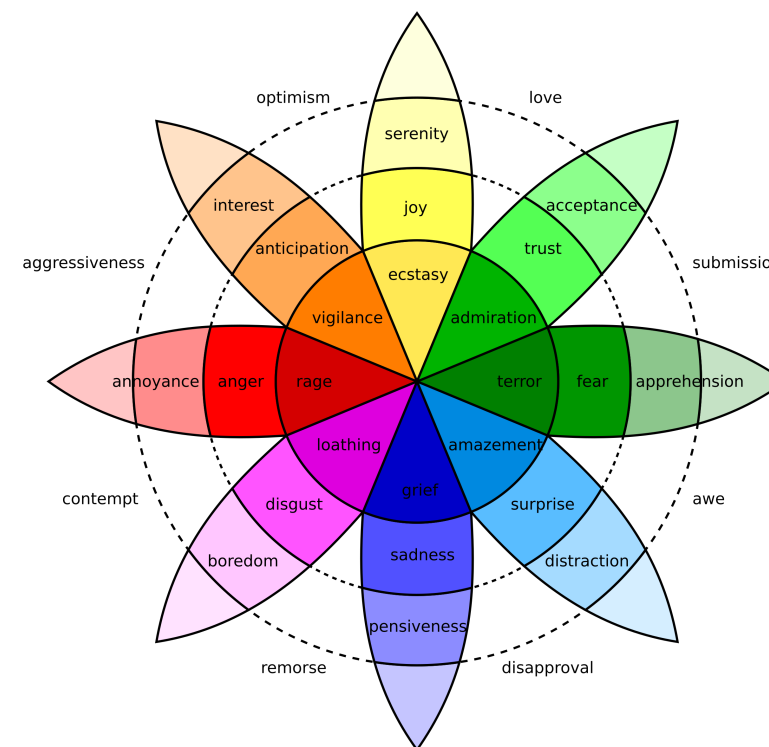
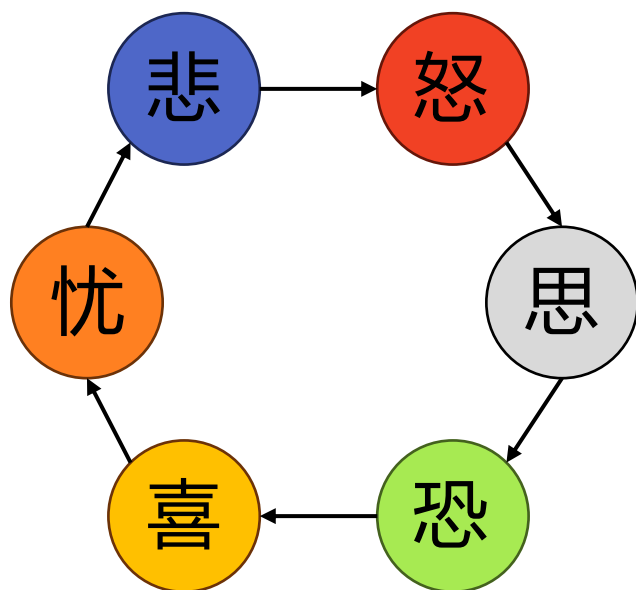
- American psychologist Robert Plutchik's "wheel of emotions"
- 4 pairs of a positive or negative basis:
 - joy vs. sadness
 - anger vs. fear
 - trust vs. disgust
 - surprise vs. anticipation



区别与联系

Source: <https://ctext.org/huangdi-neijing/yin-yang-ying-xiang-da-lun/zhs>

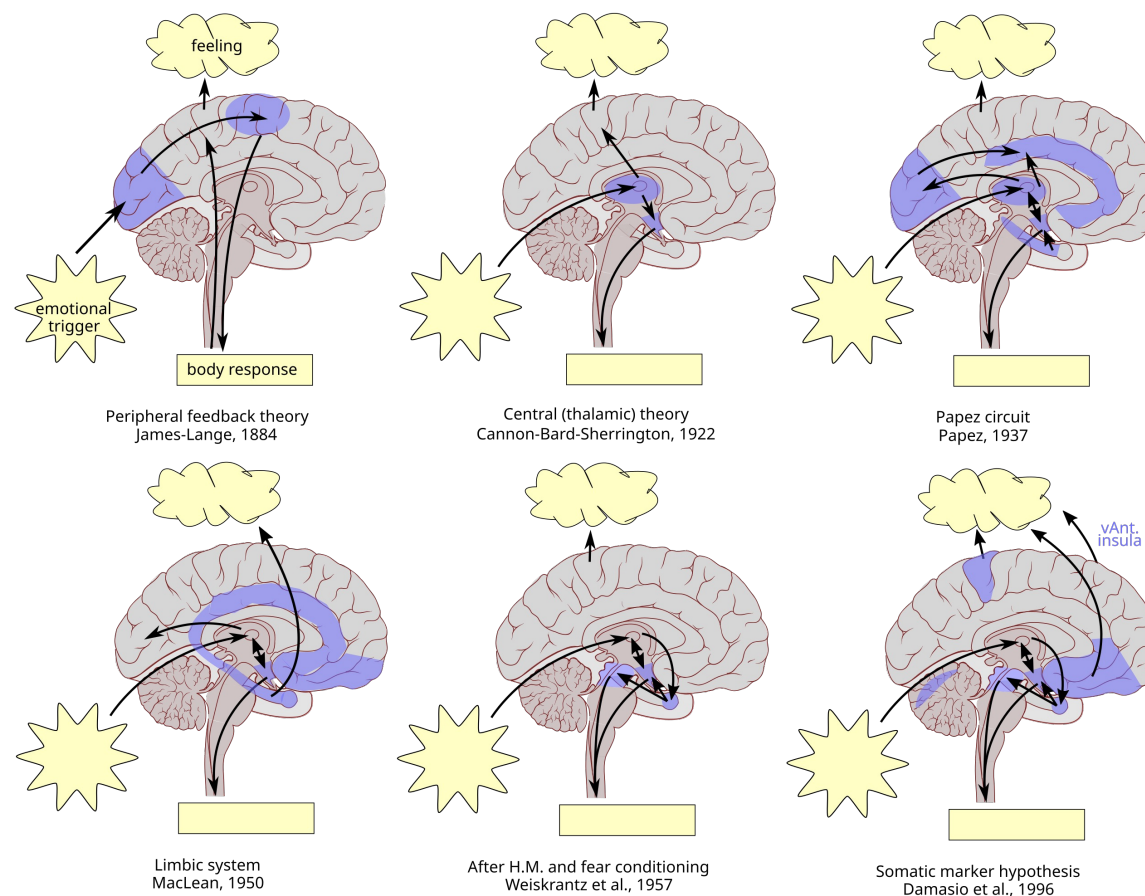
- 《黄帝内经·素问》：悲胜怒.....怒胜思.....思胜恐.....恐胜喜.....喜胜忧.....



情绪与生理

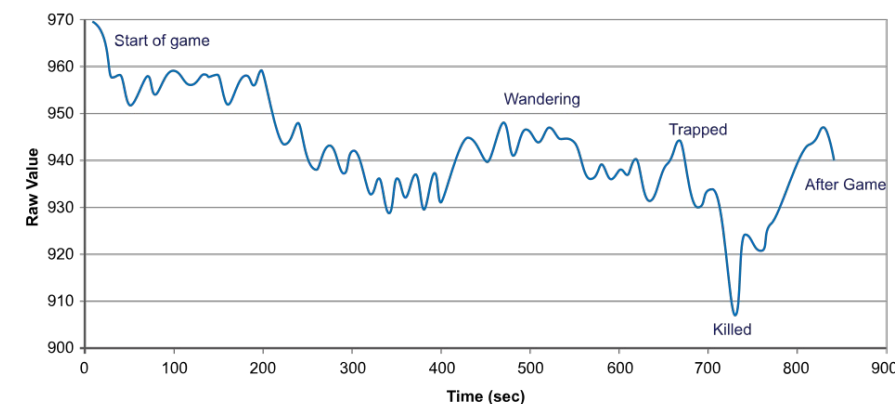
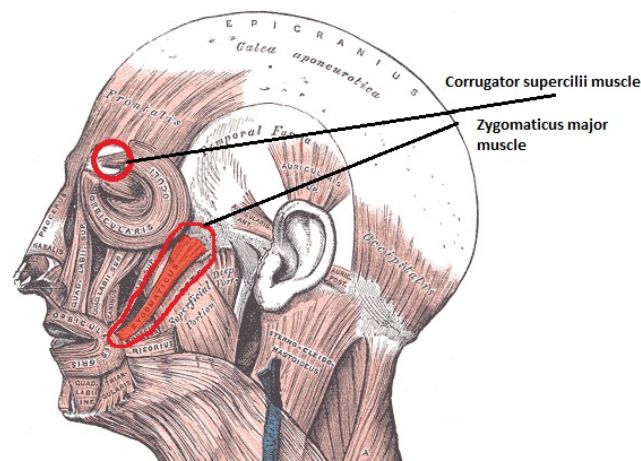
Source: <https://en.wikipedia.org/wiki/Emotion>

- 神经机制：情绪是哺乳动物大脑边缘系统组织结构（包括下丘脑 Hypothalamus、扣带回 Cingulate gyrus、海马体 hippocampus）的一种愉悦或不愉悦的心理状态。（与爬行动物有区别）
- 伴随神经化学物质的分泌，增强或减弱大脑的活动水平。



Affective Computation

- Affective Computation 情感计算：旨在研发能够识别、解释、处理、模拟人类情感的系统。
- 多模态技术：语音情感、面部情感、身体姿势、生理检测 (皮肤电位等)、视觉审美.....



玩游戏过程中皮肤电位的变化曲线

Emotion in LLMs

Source: <https://aclanthology.org/2024.acl-long.326.pdf>

- Basic capabilities: Emotion recognition.
- Advanced: emotion regulation and thought facilitation.

Understand emotions via reasoning & Imply causes from the context

